Neural network mechanics

Unit 2 · reference sheet · AI/ML in 35 topics

Forward pass

$z^\ell = W^\ell a^{\ell-1} + b^\ell \quad a^\ell = \varphi(z^\ell)$
$W^\ell \in \mathbb{R}^{n_\ell \times n_{\ell-1}}$ params $= n_\ell n_{\ell-1} + n_\ell$
Batch: $Z^\ell = A^{\ell-1}(W^\ell)^\top + b^\ell$, shape $B \times n_\ell$

Why the bend is mandatory

$W_2(W_1 x + b_1) + b_2 = (W_2 W_1)x + (W_2 b_1 + b_2)$
$\Rightarrow$ any depth collapses to one matrix without φ.

Backprop

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial \text{out}} \cdot \frac{\partial \text{out}}{\partial z} \cdot \frac{\partial z}{\partial w}$$

Fan-out: gradients **add**. Reverse topological order, one sweep.

Training loop

```
pred = model(x); loss = crit(pred,y)
loss.backward(); opt.step(); opt.zero_grad()
```

steps = epochs $\times N/B$ horizon: lr low $\rightarrow$ plateau

Loss curve decoder

Shape	Cause	Action
flat at $-\log(1/C)$	uniform output	lr, labels, zero_grad
falls then rises	overfit	early stop
NaN	lr too high	$\div 10$, warmup

Gotchas

- Skip `zero_grad` $\Rightarrow$ gradients accumulate.
- Eval with dropout on $\Rightarrow$ noisy, wrong metric.
- Dead ReLU: gradient 0 forever. LeakyReLU/GELU or lower lr.

This unit at a glance

#	Topic	You can
8	The neuron	prove that stacked linear layers collapse to one linear layer, so nonlinearity
9	Layers & the forward pass	trace shapes and values through a 2-layer net by hand
10	Backpropagation	apply the chain rule to a small graph and say what each gradient means
11	The training loop	write the 5-line loop; explain learning rate, batch, epoch, and read a loss cu

If you see this $\rightarrow$ do that

Signal	Move
Add depth	activation + residual + norm
Loss NaN	clip, lower lr
Unit outputs 0 always	LeakyReLU / lower lr
Params explode	weight sharing
Output must be a distribution	softmax

Numbers worth memorising

Params $512 \rightarrow 512$: 262,656
2-layer net $2 \rightarrow 2 \rightarrow 1$: 9
Dead network: output $= \sigma(b_2)$, independent of input
$-\ln(0.5) = 0.693$: loss of a uniform binary model